

Karim Almomani

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Portfolio: <https://themomani.github.io/Portfolio/>

Computer Engineering student interested in embedded systems and firmware development, with experience in ARM Cortex-M microcontrollers, embedded C, and the integration of hardware and software.

EDUCATION

McMaster University

B.Eng. in Computer Engineering, GPA: 3.80/4.00

Hamilton, Canada

Expected Graduation: 2029

EXPERIENCE

EcoCAR McMaster – EV Design Competition Team

Systems Design & Integration Team Member

Hamilton, Canada

Oct 2025 – Present

- High Voltage subteam member on an EV design competition team, which came in **second place in 2026**
- Co-presented a **PCB design workshop** in **Altium Designer**
- Performed hands-on inspection of **automotive inverters** and High Voltage Cable maintenance
- Documented and carried out High Voltage safety procedures
- Performed soldering tasks on PCBs for the Vehicle Inspection subteam

Orange Telecommunications

Software Development Intern

Amman, Jordan

May 2025 – Aug 2025

- Developed responsive web pages integrating **APIs**, HTML5, CSS, Bootstrap, and JavaScript
- **Mentored** incoming interns by setting onboarding goals and providing ongoing technical feedback

PROJECTS

3D Environment Scanner

2DX4: MCU Course Final Project

Winter 2026

- Engineered **Embedded C firmware** on an **ARM Cortex-M4** (TI MSP432E401Y) to rotate a VL53L1X Time-of-Flight sensor via **I2C** and a **GPIO**-controlled stepper motor, taking multiple 360° scans and joining them to reconstruct a live 3D model
- Implemented **polling** for sensing button inputs (scan/pause/return), & **UART** (115200 baud) sending CSV data to the PC
- Built a **Python** program (**PySerial**, **Matplotlib**) to parse serial data, flag anomalies, and create a 3D render of surroundings
- Debugged UART using **RealTerm**, firmware execution with **Keil Debugger**, and Python visualization using test datasets

Digital Logic Design Labs

2DI4: Logic Design

Fall 2025

- Designed and implemented finite state machines on **FPGA** boards using **Intel Quartus**
- Designed simple RAM modules as part of laboratory and coursework

EV Charger Finder

Sole Developer

Summer 2025

- Developed a program allowing users to find EV chargers which are compatible with their cars, within a selected city
- Integrated APIs and the leaflet.js map library

3-Bit Digital-To-Analogue Converter

2EI4: Bonus Electronics Project

Winter 2026

- Designed and simulated (**LTSpice**) a **3-bit binary-weighted-resistor DAC** with an **inverting summing op-amp**, achieving 0.714 V/step over 0-5V, while limiting **input current draw to <10 nA** by utilizing buffer and comparator op-amps
- Tested performance using **Digilent Analog Discovery 2's** oscilloscope, measuring **-1% gain error**

ZenLenZ (Phone Stabilization System)

Design Lead

MedSprint Hackathon

March 2025

- Designed a joystick-controlled camera stabilizer to aid users suffering from ALS in taking pictures with their phones
- Took complete responsibility for the CAD and 3D printed it, acting as the frame and axles for the device

Snake Game

2SH4: C++ Course Project

Fall 2025

- Developed a CLI Snake game in C++ using pointers and dynamic memory allocation
- Implemented core game logic, such as input handling and memory management

SKILLS

Programming: C, C++, Python, JavaScript

Embedded Systems: Embedded C, ARM Cortex-M, UART, I2C, FPGA

Tools: Keil uVision, LTSpice, Altium Designer, Intel Quartus, Digilent WaveForms, RealTerm, Git, GitHub, Autodesk Inventor (CAD)